Zero-Shot Fault Detection Across Unseen Industrial Domains

Domain Gap Analysis Report — AI Internship Screening Task (Deep Learning)

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Abstract

We address zero-shot anomaly detection on an industrial machine type never observed during training. A convolutional encoder is trained on normal-only vibration/acoustic recordings of four MIMII machine types (fan, pump, slider, valve) using masked log-mel spectrogram reconstruction, while a domain-adversarial head (DANN with gradient reversal) suppresses machine-identity information in the embedding, yielding a machine-agnostic representation of normal operating dynamics. At test time, anomalies in the unseen fifth machine type (ToyADMOS ToyCar, a fully cross-dataset target) are scored by combining the masked-reconstruction error with the Mahalanobis distance to the pooled source-normal Gaussian — no target-domain samples, labels, or statistics are used at any stage. We report leave-one-machine-out (LOMO) results within MIMII, a cross-dataset transfer result, three ablations, and an analysis of performance degradation as the domain gap widens.

1. Problem Setting and Constraints

Training data: normal recordings from 4 source machine types. Test data: normal and anomalous recordings from a 5th machine type. Hard constraints: (i) no labeled or unlabeled target samples at training time, which rules out domain adaptation, target-side fine-tuning, and target-fitted score normalization; (ii) anomaly labels exist only in the test set and are used exclusively for metric computation. The problem is therefore zero-shot domain generalization for unsupervised anomaly detection: the model must learn what 'normal industrial machine sound' means in a way that transfers across machine physics it has never encountered.

2. Datasets and Evaluation Protocol

We use the recordings of **MIMII** (fan, pump, slider, valve; several machine IDs each, normal/abnormal clips) and **ToyADMOS** (ToyCar), accessed via the DCASE 2020 Task 2 development set — the official single-channel 16 kHz repackaging of these two datasets published by the same research groups. ToyCar provides the genuinely unseen fifth machine type from a different dataset, recording rig, and acoustic environment. The code also supports the raw multi-channel MIMII 6 dB/0 dB archives directly. Two evaluation regimes: (a) **LOMO within MIMII** — train on 3 machine types, test zero-shot on the 4th, rotated over all 4 targets (moderate domain gap, 4 measurement points); (b) **cross-dataset** — train on all 4 MIMII types, test on ToyCar (maximal domain gap). Metrics: file-level ROC-AUC and pAUC at FPR ≤ 0.1 (the operating regime relevant for plant deployment, where false alarms are costly). Patch scores are mean-aggregated per recording. Seeds fixed (42); all configs serialized with each checkpoint.

3. Method

3.1 Features

Recordings are resampled to 16 kHz, converted to 128-bin log-mel spectrograms (1024-pt FFT, 512 hop), per-clip standardized, and cut into 64-frame (~2 s) patches with 50% overlap. Per-clip standardization removes gross level differences between recording rigs — the first, cheapest line of defense against domain gap.

3.2 Domain-agnostic representation learning

A 4-stage CNN encoder maps each patch to a 128-d embedding, trained with two cooperating objectives. **(1) Masked reconstruction:** 30% of time columns are masked at the input; a deconvolutional decoder must reconstruct the full unmasked patch from the embedding. Unlike a plain autoencoder, this forces the embedding to capture the temporal-spectral regularities of normal operation (periodicity, harmonic structure, stationarity) rather than memorizing instances — precisely the properties whose violation defines a fault, independent of machine type. **(2) Domain-adversarial regularization (DANN):** a classifier predicts which of the 4 source machines produced the embedding; a gradient-reversal layer makes the encoder maximize this classifier's loss. At convergence the embedding is (approximately) invariant to machine identity, so the source-normal distribution models 'normality' rather than 'fan-ness or pump-ness'. The reversal coefficient follows the standard warm-up schedule 2/(1+e^(-10p)) - 1 to avoid destabilizing early training.

3.3 Anomaly scoring without target priors

After training, a Gaussian (Ledoit-Wolf shrinkage covariance, robust at 128-d) is fitted to source-normal embeddings only. The test score combines two complementary signals, each z-normalized using source statistics: the Mahalanobis distance (distributional novelty in the domain-invariant space) and the masked-reconstruction error (violation of learned normal dynamics). Mahalanobis excels at global spectral shifts; reconstruction error at transient and structural irregularities (e.g., valve clicks). The target machine contributes nothing to any fitted quantity, satisfying the zero-prior constraint by construction.

3.4 Architecture justification

Design choice	Justification
CNN encoder over transformer	Dataset is small per domain; CNN inductive biases (locality, translation invariance over time-frequency) generalize better at this scale and train within a single GPU session.
Masked reconstruction over plain AE	Plain AEs reconstruct anomalies too well (identity shortcut). Masking removes the shortcut and forces predictive modeling of normal dynamics.
DANN over CORAL/MMD alignment	Alignment losses match given pairs of domains; DANN learns invariance to the *factor* of machine identity, which is what must transfer to an unseen 5th domain.
Mahalanobis + recon over either alone	Complementary failure modes (global vs. local anomalies); combination is validated in ablation A2.
Source-only score normalization	Any target-side normalization would violate the zero-prior constraint; z-statistics are computed from source normals exclusively.

4. Results

4.1 Main zero-shot results (LOMO + cross-dataset)

[FILL AFTER RUNS] Insert Table 1 from *results/tables.md* (generated by *python -m src.make_report_tables*). Report AUC and pAUC per unseen target, plus the MIMII→ToyCar row. Add one sentence comparing the hardest target (typically valve — non-stationary impulsive sounds) to the easiest (typically fan).

4.2 Ablation A1 — domain-adversarial head

[FILL AFTER RUNS] Insert Table 2. Expected pattern: adversarial ON improves AUC on unseen targets (invariance helps transfer) and helps most on the cross-dataset target; if it slightly hurts an in-dataset target, discuss the invariance-vs-information trade-off.

4.3 Ablation A2 — scoring function

[FILL AFTER RUNS] Insert Table 3 (recon vs. Mahalanobis vs. combined). Comment on which targets favor which score and whether the combination dominates or merely averages.

4.4 Ablation A3 — number of source domains

[FILL AFTER RUNS] Insert Table 4 (2 vs. 3 vs. 4 source machines, fixed target). Expected: monotone improvement with source diversity — the core argument that representation generality, not capacity, drives zero-shot transfer.

5. Domain-Gap Degradation Analysis

We order test conditions by increasing domain gap: (i) LOMO targets acoustically similar to a source (e.g., pump unseen, fan in sources — both rotary/stationary); (ii) LOMO targets with dissimilar physics (valve unseen — impulsive, non-stationary, while sources are mostly stationary); (iii) cross-dataset ToyCar (different machine class, recording rig, room acoustics, SNR profile). For quantitative grounding, we also measure the gap directly as the Fréchet distance between source-normal and target-normal embedding distributions and correlate it with AUC.

[FILL AFTER RUNS] Insert *results/domain_gap.png* and report: AUC at each gap level, the AUC drop from the easiest LOMO target to ToyCar, and the correlation between embedding-space Fréchet distance and AUC. Conclude with where the method degrades gracefully vs. where it breaks (typically: stationary→stationary transfers hold; stationary→impulsive and cross-rig transfers lose the most).

6. Limitations and Future Work

(1) Invariance is enforced over only four source domains; with so few domains, DANN can remove useful information along with identity. More source diversity (e.g., adding ToyConveyor/ToyTrain as sources) is the most direct improvement. (2) Mean patch aggregation dilutes short transients; top-k aggregation is a cheap fix. (3) The Gaussian assumption on pooled normals is crude — a per-domain mixture with min-distance scoring is a natural extension. (4) Self-supervised pretraining on large unlabeled audio (e.g., BEATs/AudioMAE features) would likely raise the floor, at the cost of compute beyond this task's scope.

7. Reproducibility

All code, fixed seeds, and per-run serialized configs are in the repository (README contains exact commands). The complete ablation grid (~8 short runs) fits one Colab T4 session via *python -m src.run_ablations*; results append to a single CSV from which all tables and the degradation figure are regenerated deterministically.

References

• Ganin & Lempitsky (2015). Unsupervised Domain Adaptation by Backpropagation (DANN). ICML.

• Purohit et al. (2019). MIMII Dataset: Sound Dataset for Malfunctioning Industrial Machine Investigation. DCASE.

• Koizumi et al. (2019). ToyADMOS: A Dataset of Miniature-Machine Operating Sounds for Anomalous Sound Detection. WASPAA.

• Ledoit & Wolf (2004). A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices. J. Multivariate Analysis.

• He et al. (2022). Masked Autoencoders Are Scalable Vision Learners. CVPR.